PROJECT: TICKET ASSESSMENT SYSTEM USING SERVICENOW

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1. INTRODUCTION

1.1 Project Overview

The Ticket Assessment System is developed using the ServiceNow platform to automate the process of managing and tracking service requests within an organization. In many organizations, employees and users often face technical issues or require support for different services such as software access, hardware problems, or system maintenance. Traditionally, these issues are reported through emails, phone calls, or manual records, which makes it difficult to track the progress of requests and manage them efficiently.

Manual handling of service requests can lead to several problems such as delays in issue resolution, duplication of requests, and lack of proper documentation. When the number of service requests increases, it becomes difficult for support teams to monitor and manage them effectively. This may result in poor communication between users and administrators and can reduce the overall productivity of the organization.

1.2 Purpose of the Project

The main purpose of the Ticket Assessment System is to automate the process of managing service requests and technical issues in an organization. Instead of using manual methods such as emails or spreadsheets, the system provides a centralized platform where users can submit their issues and administrators can track and resolve them efficiently.

The project aims to simplify the workflow involved in handling service requests. Users can easily create tickets by filling out a form, and the system stores the information in a structured database. Administrators can then review the tickets, assign priorities, and update the status as the issue progresses.

2.IDEATION PHASE

2.1 Brainstorming – Idea Generation – Prioritization

Brainstorming is the process of generating different ideas and identifying possible solutions for a specific problem. In this phase, several ideas were discussed to improve the process of managing service requests in organizations. Many companies experience difficulties when handling large numbers of service requests manually, and this often leads to delays and confusion in issue management.

During the brainstorming session, different solutions were considered to address the problem of inefficient request management. Some of the ideas included developing a manual database system for storing service requests, creating a simple web-based application for tracking issues, and implementing a fully automated ticket management system using a cloud platform. Each idea was carefully evaluated based on feasibility, efficiency, and ease of implementation.

2.2 Define Problem Statement

In many organizations, service requests and technical issues are still managed manually through emails, phone calls, or spreadsheets. This traditional method of handling requests is time-consuming and inefficient. As the number of users and service requests increases, it becomes difficult for support teams to track and manage all the issues effectively.

Manual management of service requests often results in several problems such as delays in resolving issues, duplication of requests, and difficulty in monitoring the progress of each ticket. Additionally, important information may be lost or miscommunicated when requests are handled without a structured system.

Another challenge is the lack of proper documentation and tracking. Without a centralized system, administrators cannot easily determine which issues are pending, which have been resolved, and which require urgent attention. This lack of visibility can negatively affect the efficiency of support services and reduce user satisfaction.

2.3 Empathy Map Canvas

The empathy map canvas is a tool used to understand the needs, experiences, and challenges of users interacting with the system. In the Ticket Assessment System, the primary users are employees, customers, and administrators who are involved in reporting or resolving service requests.

Says:
Users often say that reporting issues manually through emails or phone calls is inconvenient and time-consuming. They prefer a simple system where they can quickly submit their problems and track the status of their requests.

Thinks:
Users think that an automated ticket system would make it easier to report issues and monitor the progress of their requests. They expect the system to provide quick responses and clear updates regarding their tickets.

Does:
Currently, users report their problems through traditional communication methods such as email, messaging platforms, or phone calls. Support teams manually record these requests and attempt to resolve them.

Feels:
Users often feel frustrated when their issues are not addressed quickly or when they do not receive updates about the status of their requests. An organized ticket system can reduce this frustration by providing transparency and faster service.

By analyzing these user perspectives, the Ticket Assessment System is designed to provide a user-friendly platform that simplifies the process of reporting and managing service requests.

3. REQUIREMENT ANALYSIS

3.1 Data Flow Diagrams and User Stories

A Data Flow Diagram (DFD) is used to represent how data moves through a system and how it is processed. It helps in understanding the interaction between users and the system components. In the Ticket Assessment System, the Data Flow Diagram shows how service requests are created, processed, and managed within the system.

The process begins when a user submits a ticket through a form provided in the ServiceNow platform. The form allows the user to enter details such as the issue description, department, priority level, and other necessary information. Once the ticket is submitted, the system processes the information and stores it in a database table.

After the ticket is stored, administrators or support staff can access the database to review the submitted requests. The administrator then assesses the ticket, assigns the appropriate priority, and updates the status of the request. The system may also send notifications or updates to users regarding the progress of their tickets.

The main components of the Data Flow Diagram include:

Input:
Ticket details entered by the user through the ServiceNow form.

Process:
System processing and assessment of ticket information.

Storage:
ServiceNow database table where ticket records are stored.

Output:

Updated ticket status and notifications sent to users or administrators.

User stories help describe how different users interact with the system and what they expect from it.

Some user stories for the Ticket Assessment System include:

- As a user, I want to submit a ticket so that my issue can be reported and resolved.

- As an administrator, I want to view all submitted tickets so that I can manage service requests efficiently.

- As a support team member, I want to update the ticket status so that users can track the progress of their requests.

- As a user, I want to receive notifications when my ticket is updated or resolved.

These user stories help developers understand the system requirements and guide the development process.

3.2 Solution Requirements

Solution requirements define the features and characteristics that the system must have to operate effectively. These requirements are generally divided into functional requirements and non-functional requirements.

Functional Requirements

Functional requirements describe the main functions that the system must perform.

- The system should allow users to create and submit tickets through a form.

- The system should store ticket details in a database table.
- The system should allow administrators to view and manage all submitted tickets.
- The system should allow administrators to update the ticket status.
- The system should send notifications when ticket status changes.

Non-Functional Requirements

Non-functional requirements describe the quality and performance of the system.

- The system should be easy to use and user-friendly.
- The system should provide secure storage for ticket information.
- The system should respond quickly to user requests.
- The system should ensure reliability and data accuracy.
- The system should be accessible through the internet using the ServiceNow platform.

These requirements ensure that the Ticket Assessment System functions properly and meets the needs of both users and administrators.

3.3 Technology Stack

The Ticket Assessment System is developed using the ServiceNow platform, which is widely used for enterprise service management and workflow automation. ServiceNow provides a cloud-based environment that allows developers to create applications efficiently.

The key technologies used in this project include:

ServiceNow Developer Instance
This environment is used to develop and test the application.

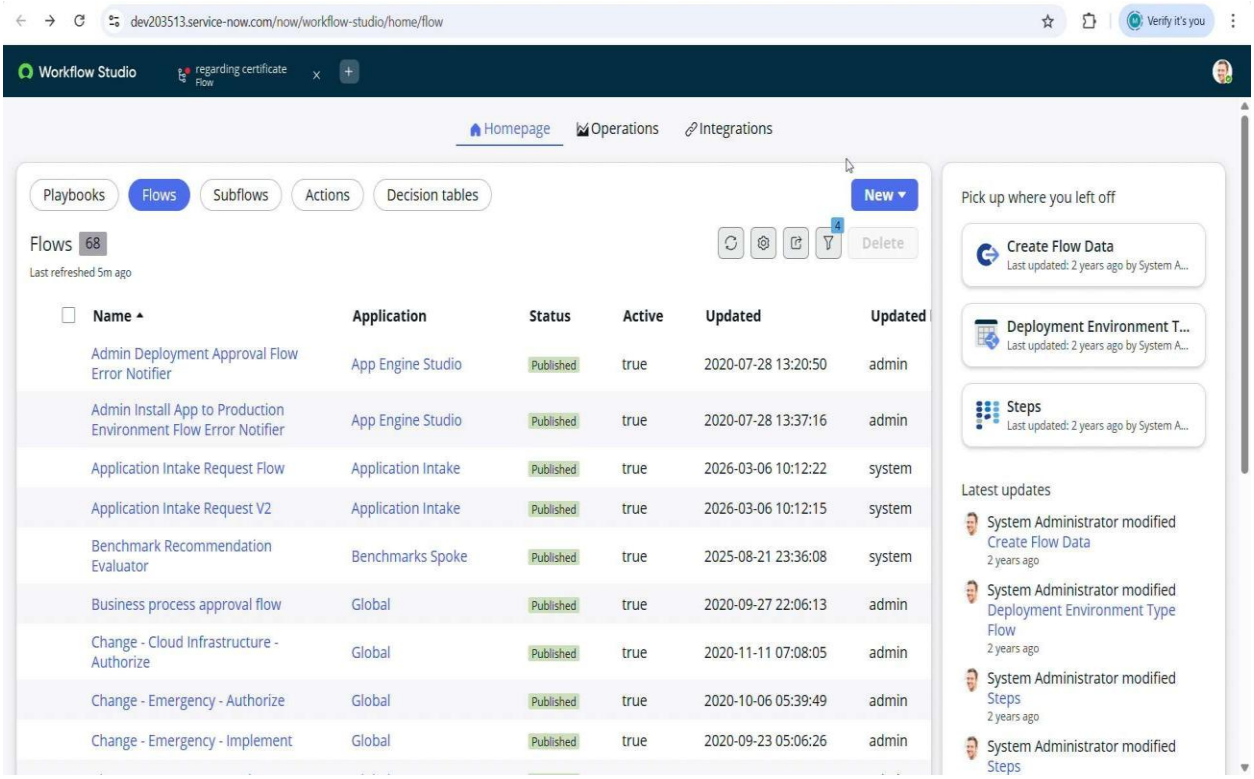
ServiceNow Tables
Tables act as the database where ticket information is stored. Each record represents a ticket submitted by a user.

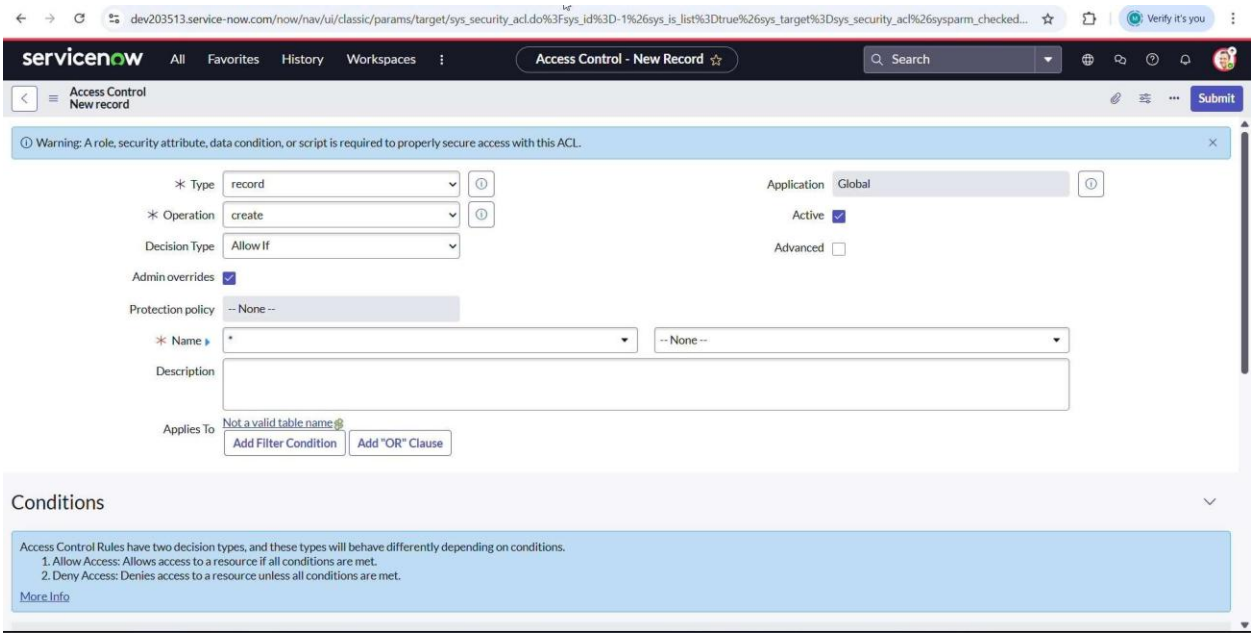
Forms
Forms are used to collect ticket details from users. These forms allow users to enter information such as issue description, department, and priority.

Workflow / Flow Designer
Workflow automation tools are used to process ticket information and perform actions automatically, such as updating ticket status or sending notifications.

Notification System
Notifications inform users and administrators about updates or changes in ticket status.

Together, these technologies create a reliable and efficient ticket management system that automates the process of handling service requests.





4.PROJECT DESIGN PHASE

4.1 Problem – Solution Fit

In many organizations, service requests and technical issues are managed manually through emails, phone calls, or spreadsheets. This traditional method is inefficient and often leads to delays in resolving problems. When the number of service requests increases, it becomes difficult for support teams to track and manage each request properly. Important information may be lost, duplicated, or delayed due to the lack of a structured system.

Another challenge is the absence of a centralized platform where all service requests can be recorded and monitored. Without such a system, administrators cannot easily determine the status of each request or prioritize issues based on urgency. This reduces the efficiency of support services and can negatively affect user satisfaction.

The Ticket Assessment System solves this problem by providing an automated platform where users can submit tickets describing their issues. The system records the ticket information in a database and allows administrators to review and manage the requests effectively. Automation features ensure that tickets are processed systematically and that updates are communicated to the users.

By implementing this solution, organizations can improve the efficiency of service request management. The system reduces manual work, improves communication between users and administrators, and ensures that all service requests are properly documented and tracked.

4.2 Proposed Solution

The proposed solution is to develop a Ticket Assessment System using the ServiceNow platform. This system provides a centralized and automated environment for managing service requests within an organization.

In this system, users can submit their issues through a digital form. The form collects necessary information such as user details, issue description, department, and priority level. Once the form is submitted, the system automatically stores the ticket information in a database table.

4.3 Solution Architecture

The solution architecture describes how different components of the Ticket Assessment System interact with each other. It explains the structure of the system and the flow of information between users, the system, and administrators.

The architecture of the system consists of several main components.

First, the User Interface allows users to interact with the system. Users submit their service requests by filling out a ticket form available on the ServiceNow platform.

Second, the ServiceNow Database Tables store all ticket information. Each ticket created by a user is recorded as a new entry in the database table. The stored information includes details such as ticket ID, user name, issue description, priority level, and status.

Third, the Workflow Automation component processes ticket information automatically. Workflows may send notifications, update ticket status, or trigger other actions depending on the system rules.

Finally, the Administrator Interface allows administrators or support staff to access the system and manage service requests. They can view submitted tickets, update the ticket status, assign tasks, and resolve issues

5. PROJECT PLANNING PHASE

5.1 Project Planning

Project planning is an important phase in the development of the Ticket Assessment System. It involves identifying the tasks required to complete the project and organizing them in a structured manner. Proper planning helps ensure that the project is completed efficiently within the given time frame.

The planning process begins with defining the project objectives. The main objective of this project is to design and implement a ticket management system that allows users to submit service requests and enables administrators to manage those requests effectively.

After defining the objectives, the project tasks are identified. These tasks include requirement analysis, system design, development, testing, and documentation. Each task is assigned a specific timeline to ensure that the project progresses in an organized manner.

Another important aspect of project planning is resource management. The system is developed using the ServiceNow platform, which provides tools for creating forms, workflows, and database tables. Proper utilization of these tools helps simplify the development process.

Finally, the project schedule is created to track the progress of different phases. Regular monitoring ensures that any issues or delays can be addressed promptly. This structured approach helps ensure the successful completion of the project.

5.2 Project Timeline

The project timeline provides a clear overview of the different stages involved in the development of the Ticket Assessment System. Each stage of the project is completed within a specific period to ensure systematic progress.

The first stage of the project is the Ideation Phase, where the problem is identified and possible solutions are explored. During this phase, tools such as empathy maps and brainstorming techniques are used to understand user needs.

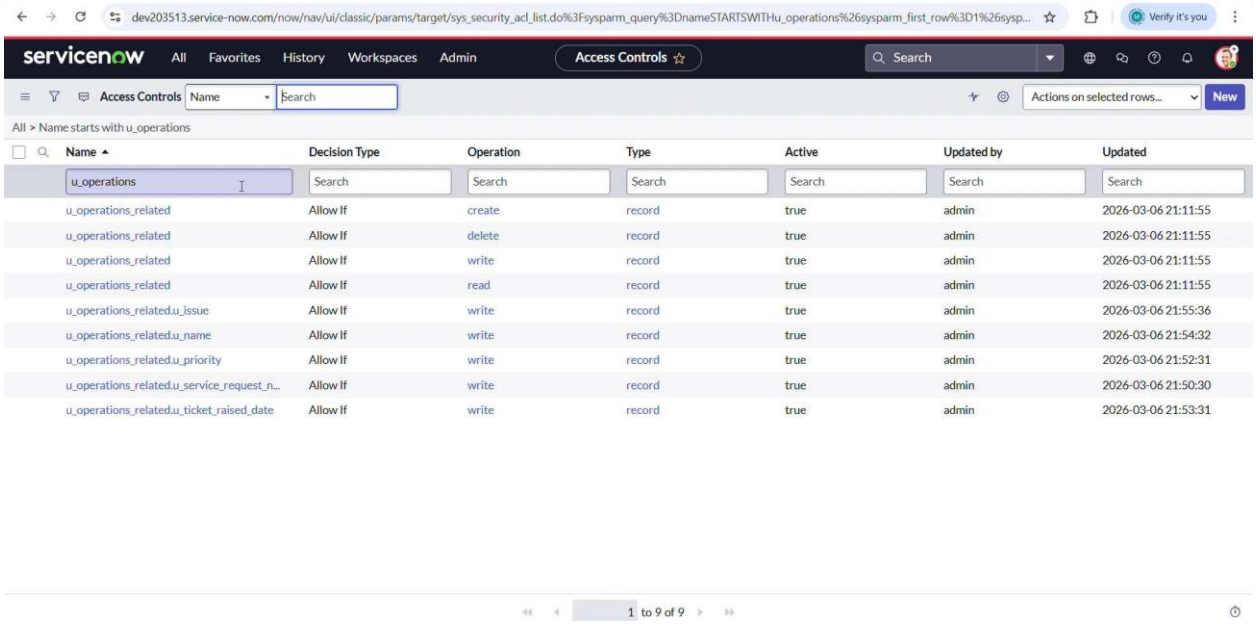
The second stage is the Requirement Analysis Phase, where the system requirements are defined. In this phase, functional and non-functional requirements are identified to guide the development process.

The third stage is the Project Design Phase, where the system architecture and solution approach are planned. This phase includes designing the workflow, database structure, and user interface.

The fourth stage is the Development Phase, where the actual system is built using the ServiceNow platform. Forms, tables, and workflows are created to implement the system functionality.

The fifth stage is the Testing Phase, where the system is tested to ensure that all features work correctly. Errors or bugs identified during testing are fixed before final deployment.

The final stage is the Documentation and Deployment Phase, where the project report is prepared and the system is made ready for practical use.



6. PROJECT DEVELOPMENT PHASE

6.1 System Development

The Project Development Phase is one of the most important stages in the creation of the Ticket Assessment System. In this phase, the design and requirements identified in earlier stages are implemented into a functional system. The system is developed using the ServiceNow platform, which provides a cloud-based environment for building applications and managing workflows.

The development process begins with the creation of a custom application within the ServiceNow platform. This application serves as the main environment where all the components of the Ticket Assessment System are built and managed. The application includes database tables, forms, workflows, and user interfaces.

The next step involves creating database tables to store ticket information. These tables contain various fields such as ticket ID, user name, department, issue description, priority level, and ticket status. Each ticket submitted by a user is stored as a record in the table, allowing administrators to access and manage the information easily.

6.2 System Implementation

System implementation involves deploying the developed Ticket Assessment System so that users and administrators can start using it in a real environment. During this stage, the application created on the ServiceNow platform is configured and prepared for actual use.

The implementation process begins with testing the application features to ensure that all components are functioning properly. This includes verifying that ticket submission forms work correctly, data is stored in the database tables, and workflows perform the required automated actions.

7. ADVANTAGES AND DISADVANTAGES

7.1 Advantages

The Ticket Assessment System offers several benefits for organizations that need to manage service requests efficiently. One of the main advantages is centralized ticket management. All service requests are stored in a single system, which makes it easier for administrators to track and manage them.

Another advantage is improved communication between users and support teams. Users can submit their issues through the system and receive updates on the progress of their tickets. This reduces confusion and ensures that users are informed about the status of their requests.

The system also provides automation features. When a ticket is created, the system can automatically assign a ticket number and notify the support team. Automation reduces manual work and speeds up the issue resolution process.

Additionally, the system improves data organization and record keeping. All ticket information is stored securely in database tables within the ServiceNow platform, making it easy to retrieve past records for reference or analysis.

Finally, the Ticket Assessment System increases overall operational efficiency. Administrators can prioritize tickets based on urgency and ensure that critical issues are addressed quickly.

7.2 Disadvantages

Despite its advantages, the Ticket Assessment System also has some limitations. One of the main disadvantages is the initial setup and configuration effort required to implement the system. Administrators must design forms, workflows, and database tables before the system can be used effectively.

Another limitation is the dependency on internet connectivity. Since the system operates on a cloud-based platform like ServiceNow, users must have internet access to submit or manage tickets.

There may also be a learning curve for new users. Individuals who are not familiar with the system may require some training before they can use it efficiently.

Additionally, if the system is not maintained properly, it may result in outdated records or unresolved tickets. Regular monitoring and maintenance are necessary to ensure smooth operation.

Although these disadvantages exist, they can be minimized through proper system management and user training.

1. Initial Setup Complexity

Implementing the system may require time and effort during the initial setup stage. Administrators need to configure forms, workflows, and database tables before the system becomes fully operational.

2. Dependence on Internet Connectivity

Since the system operates on a cloud-based platform like ServiceNow, it requires a stable internet connection. If the internet connection is unavailable, users may not be able to submit or access tickets.

3. Training Requirements

Users and administrators may require training to understand how to use the system effectively. Individuals who are unfamiliar with the platform might initially find it difficult to navigate the interface.

8. PROJECT DOCUMENTATION

Project documentation provides a detailed explanation of the system design, functionality, and implementation. It serves as a reference for understanding how the Ticket Assessment System was developed and how it operates.

The documentation includes information about the system architecture, workflow processes, and database structure. It also explains the steps involved in creating the ticket submission form and managing service requests through the system.

Another important part of the documentation is the screen layout section, which includes screenshots of the system interface. These screenshots help demonstrate how users interact with the system and how administrators manage tickets.

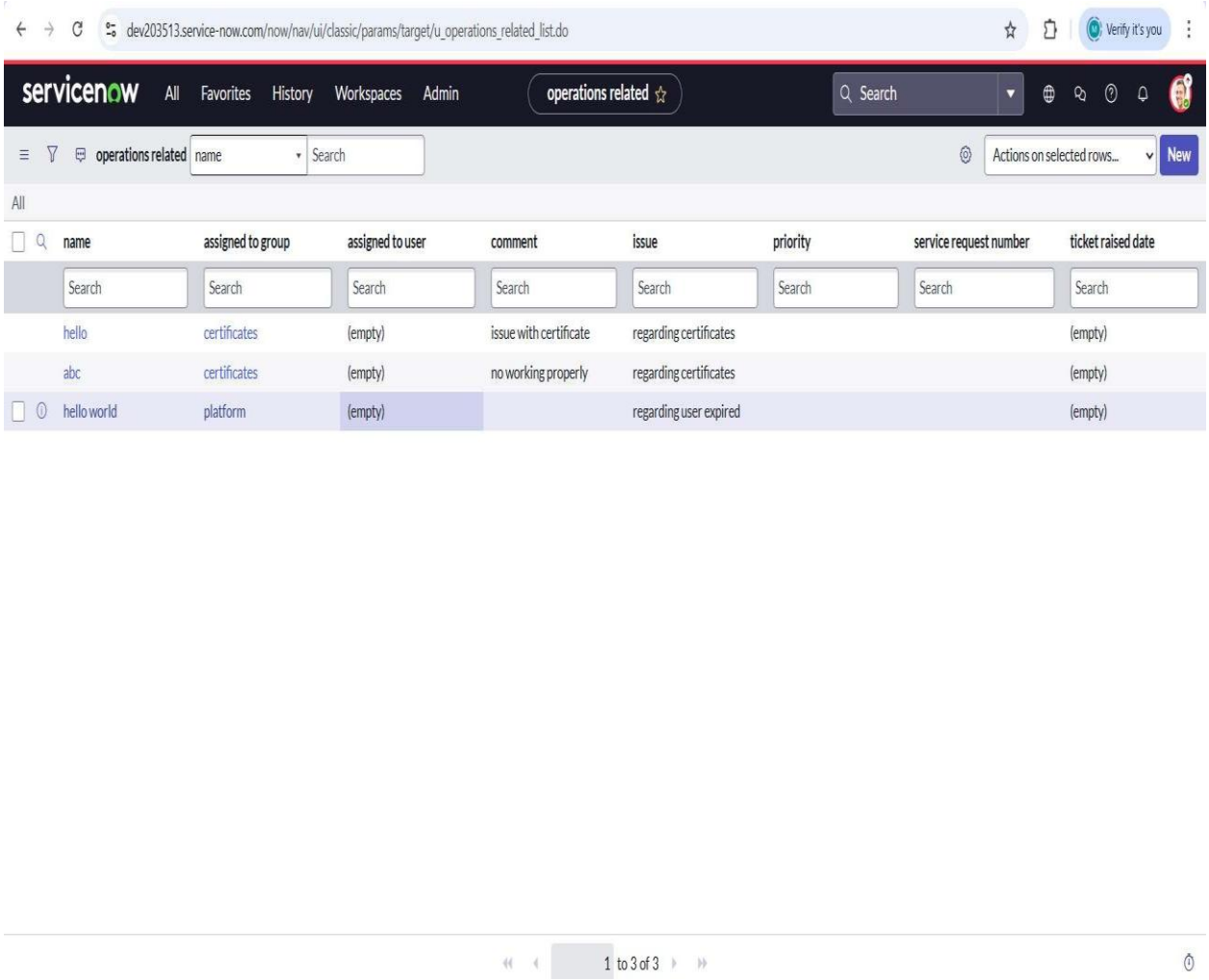
Examples of screen layouts included in the documentation are:

- Login page of the system
- Ticket submission form
- Ticket list view
- Ticket detail page
- Administrator dashboard
- Workflow configuration page

These screenshots provide a visual representation of the system and make it easier to understand its functionality.

The project documentation also includes explanations of system features, testing results, and implementation details. Proper documentation ensures that the system can be easily maintained and improved in the future.

OUTPUT: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9.CONCLUSION

The Ticket Assessment System is designed to improve the process of managing service requests within an organization. Traditional methods of handling service issues, such as emails or manual records, can be inefficient and difficult to track. The implementation of an automated ticket management system helps address these challenges.

By using the ServiceNow platform, the system provides a centralized environment where users can submit their issues and administrators can manage them effectively. The system organizes service requests in a structured format, making it easier to monitor and resolve problems.

Throughout the development process, different phases such as ideation, requirement analysis, design, development, and testing were completed to ensure that the system meets user needs. Each phase contributed to building a reliable and efficient solution.

The Ticket Assessment System improves communication between users and support teams, enhances record management, and reduces manual work through automation. These benefits help organizations provide faster and more efficient support services.

In the future, additional features such as advanced reporting, automated prioritization, and mobile accessibility can be added to further improve the system. Overall, the project demonstrates how technology can be used to streamline service request management and improve operational efficiency.